Unit #2
Chapter #16.2: Digital Logic Inverters
Lecture 7

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by Sedra and Smith
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Example 16.2a

For the simple NMOS inverter, derive expressions for V_{OH}, V_{OL}, V_{IL}, V_{IH}, and V_M. Neglect λ and show parameters can be expressed in terms of V_{DD}, V_t, and $k_nR_D=1/V_x$

Example 16.2a Cont.

For the simple NMOS inverter, derive expressions for V_{OH}, V_{OL}, V_{IL}, V_{IH}, and V_M. Neglect λ and show parameters can be expressed in terms of V_{DD}, V_t, and $k_nR_D=1/V_x$

Example 16.2a Cont.

For the simple NMOS inverter, derive expressions for V_{OH}, V_{OL}, V_{IL}, V_{IH}, and V_M. Neglect λ and show parameters can be expressed in terms of V_{DD}, V_t, and $k_nR_D=1/V_x$

Example 16.2a Cont.

For the simple NMOS inverter, derive expressions for V_{OH}, V_{OL}, V_{IL}, V_{IH}, and V_M. Neglect λ and show parameters can be expressed in terms of V_{DD}, V_t, and $k_nR_D=1/V_x$

Example 16.2c

Find numerical values for all parameters assuming V_{DD}=1.0V, V_t=0.35V, V_x = 0.0225V.

Example 16.2d

For $k_n' = 540uA/V^2$ and W/L = 1.5, find the required value of R_D and use it to determine the average power dissipated in the inverter, assuming that the inverter spends half of the time in each of its two states. Use V_{DD} = 1.0V, $V_{x} = 0.0225V$ and V_{OL} = 33 mV.